

MICROCONTROLLER-BASED ELECTRONIC CALCULATOR

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An electronic calculator is a device used to perform the basic arithmetic operations. It is used to calculate problems in science, engineering and mathematics. Here we present a microcontroller-based electronic calculator. This calculator can perform functions like addition, subtraction, multiplication, division, reciprocal, square root and exponential for positive and negative real numbers.

Circuit description

Fig. 1 shows the circuit diagram of microcontroller-based electronic calculator. Microcontroller AT89C52 is at the heart of this calculator. The AT89C52 is a low-power, high-performance CMOS 8-bit microcomputer with 8k bytes of Flash programmable and erasable read only memory (EPROM). The AT89C52 provides standard features—256 bytes of RAM, 32 input/output lines, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

System clock plays a significant role in the operation of the microcontroller. A 12MHz quartz crystal at pins 18 and 19 provides clock pulse to the microcontroller (IC1). Power-on reset is provided by electrolytic capacitor C3 and resistor R2. Switch S1 is used for manual reset. Microcontroller port pins P1.4 through P1.7 and P3.2 through P3.7 are used to form the keypad matrix. Port P2 is connected with the seven-segment display. Port P2 provides segment data to glow the LED segments of the seven-segment display. Eight

100-ohm resistors are used to limit the current through LEDs segment of seven-segment display. Port P0 is pulled high with 10k resistor network RNW1.

Microcontroller port pins P0.0 through P0.7 drive the DIS1 through DIS8 with the help of transistors T1 through T8, respectively. Microcontroller port pin P0.0 goes low to drive transistor T1 into saturation and provides supply to common-anode pin 3 of DIS1. Similarly, transistors T2 through T8 drive common-anode of seven-segment display pin 3 of DIS2 through DIS8, respectively. Microcontroller provides the segment data and display-enable signal simultaneously in time division multiplexed mode for displaying a particular number on the seven-segment display unit. Segment data and display-enable pulse for display are refreshed after every few millisecond delay. Thus the display

PARTS LIST

Semiconductors:	
IC1	- AT89C52 microcontroller
IC2	- 7805, 5V regulator
T1-T8	- BC557 pnp transistor
DIS1-DIS8	- LTS542 common anode seven-segment display
D1-D4	- 1N4007 rectifier diode
LED1	- 5mm LED
Resistors (all ¼-watt, ±5% carbon)	
R1	- 470-ohm
R2	- 10-kilo-ohm
R3-R10	- 1-kilo-ohm
R11-R18	- 100-ohm
RNW1	- 10-kilo-ohm resistor network
Capacitors:	
C1	- 1000µF, 25V electrolytic
C2	- 0.1µF ceramic disk
C3	- 10µF, 16V electrolytic
C4, C5	- 22pF ceramic disk
Miscellaneous:	
X1	- 230V AC primary to 9V, 500mA secondary transformer
X _{CRYST}	- 12MHz crystal
S1-S24	- Push-to-on tactile switch



appears to be continuous even though it lights up one by one.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC2). Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R1 limits the current through LED1.

An actual-size, single-side PCB for the microcontroller-based electronic calculator is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked errors.

Working of the calculator

The calculator works like any simple calculator available in the market. The various digits and signs can be entered using numeric keys 0 to 9 and 00, decimal point with 'dp' and '+/-' for the sign of the number. The display will show 'E', if the answer is infinite. Various functions include:

Summation.

First addend '+' second addend '='

The answer is the sum of these two addends

Subtraction.

Minuend '-' subtrahend '='

The answer is the difference of these two operands

Multiplication.